

AI CyberShield Prototype – Output Explanation

Introduction

The AI CyberShield prototype is an AI-powered cybersecurity platform that detects phishing emails, fake websites, and deepfake media. It analyzes user inputs using Artificial Intelligence, Machine Learning, Natural Language Processing, and Deep Learning to generate a real-time threat score, risk evaluation, and security recommendations.

Email Threat Detection

The system analyzes email content, sender details, suspicious links, urgency keywords, and phishing patterns. Malicious emails are flagged with a high threat score (e.g., 92%) and users are advised to quarantine the email, avoid clicking links, and report the attack. Legitimate emails receive a low threat score (e.g., 8%) and are marked as Verified Secure/Safe.

Fake Website Detection

Users enter a website URL for verification. The system checks HTTPS security, domain reputation, WHOIS details, blacklist databases, SSL certificates, domain age, and URL similarity. Suspicious websites are marked as phishing while legitimate websites are verified as safe.

Deepfake Detection Engine

Users upload images or videos. The AI analyzes facial features, eye movement, lip synchronization, frame consistency, biometric patterns, and manipulation artifacts. Authentic media is marked safe, while manipulated media is flagged as a potential deepfake.

Risk Evaluation Summary

The dashboard calculates a threat score after every scan. Green (0–30%) indicates safe content, Yellow (31–70%) indicates suspicious content, and Red (71–100%) indicates a critical threat requiring immediate action.

Dashboard Features

The dashboard includes Login & Registration, Email Threat Detection, Fake Website Detection, Deepfake Detection, AI Detection Engine, Risk Evaluation Summary, Threat Analysis, Prevention Guidelines, and Team Workspace.

Technologies Used

Frontend: HTML5, CSS3, JavaScript, Tailwind CSS. Backend: Python and FastAPI. AI/ML: TensorFlow, Scikit-learn, NLP, Deep Learning. Database: MongoDB and Firebase. APIs: VirusTotal, WHOIS, Threat Intelligence APIs.

Workflow

1. User submits an email, URL, or media file.
2. AI analyzes the input.
3. Threat patterns are identified.
4. Risk score is generated.
5. Dashboard displays findings and recommendations.
6. User takes appropriate security action.

Benefits

Protects against phishing attacks, fake websites, and deepfakes; provides real-time threat detection, improves cybersecurity awareness, reduces fraud, and offers an easy-to-use AI-powered dashboard.

Conclusion

AI CyberShield demonstrates how AI can proactively identify cyber threats and help users make safer online decisions through intelligent detection, risk scoring, and preventive guidance.